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B.Tech. Degree VI Semester Special Supplementary Examination ***January 2016***

ME 1604 HEAT AND MASS TRANSFER **(2012 scheme)**

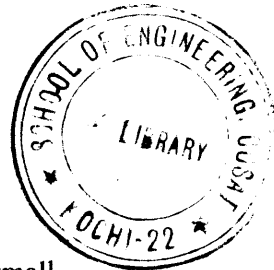
Time : 3 Hours

Maximum Marks : 100

PART A **(Answer *ALL* questions)**

(8 × 5 = 40)

- I. (a) What is meant by critical radius of insulation? Explain its significance.
- (b) What is the difference between fin effectiveness and fin efficiency?
- (c) Explain thermal and hydrodynamic boundary layers.
- (d) What is meant by critical heat flux? What is its significance?
- (e) Explain Kirchoff's law and from this law establish Kirchoff's identity.
- (f) What is a black body? What are its properties? Why does a cavity with a small hole behave as a black body?
- (g) What is meant by fouling? How does it affect the performance of heat exchangers?
- (h) What are the limitations of LMTD method? How is effectiveness – NTU method superior to LMTD method?



PART B

(4 × 15 = 60)

- II. (a) Explain the concept of overall heat transfer coefficient and derive an expression for the same for a plane wall. (5)
- (b) An exterior wall of a house may be approximated by a 5 cm layer of common brick of thermal conductivity 0.8 W/m °C, followed by a 2 cm layer of gypsum plaster of thermal conductivity 0.58 W/m °C. What thickness of loosely packed rock-wool insulation of thermal conductivity of 0.055 W/m °C should be added to reduce the heat loss (or gain) through the wall by 70%? (10)
- OR**
- III. (a) Two pin fins are identical except that the diameter of one is twice that of other. For which fin will (i) Fin effectiveness (ii) Fin efficiency be higher. (5)
- (b) Three rods of copper, aluminium and stainless steel are coated with wax all round and are dipped vertically in a water bath at 85 °C. The length of each rod projecting outside the bath is 300 mm. Diameter of each rod is 20 mm and length is 400 mm. Convective heat transfer coefficient at the surface of each rod is 11 W/m²K. Thermal conductivity of Copper rod = 380 W/mK, Aluminium rod = 206 W/mK, Steel rod = 17 W/mK. Calculate the ratio of lengths of rod, up to which wax melting occurs, due to transfer of heat. (10)

(P.T.O.)

- IV. (a) What is critical Reynolds number? State its approximate values for flow over a flat plate and through a circular tube. (5)
- (b) Air at velocity of 3 m/s and 20 °C flows over a flat plate along its length. The length, width and thickness of the plate are 100 cm, 50 cm and 2 cm respectively. The top surface of the plate is maintained at 100 °C. Calculate the heat lost by the plate and temperature of bottom surface of the plate for the steady state conditions. The thermal conductivity of the plate may be taken as 23 W/mK. (10)

OR

- V. (a) Discuss modes of condensation. Why dropwise condensation is preferred? What are the practical difficulties in retaining dropwise condensation on a surface? (6)
- (b) A vertical plate 15 cm high and 10 cm wide is maintained at 140 °C. Calculate the maximum dissipation rate from both the sides of the plate in an ambient temperature of 20 °C. The radiation heat transfer coefficient is 9.0 W/m²K. (9)
- VI. (a) Define geometrical or shape factor. What are its salient features? (7)
- (b) A cubical room 4 m by 4 m by 4 m is heated through the floor by maintaining it at uniform temperature of 350 K, while side walls are insulated. The heat loss takes place through the ceiling at 300 K. Assuming all the surfaces have an emissivity of 0.8, determine the rate of heat loss by radiation through the ceiling. (8)

OR

- VII. (a) What is a radiation shield? Where is it used? (5)
- (b) Two large parallel planes with emissivity 0.6 are at 900 K and 300 K. A radiation shield with one side polished and having emissivity of 0.05, while the emissivity of the other side is 0.4 is proposed to be used. Which side of the shield is to face the hotter plane, if the temperature of shield is to be kept minimum? Justify your answer. (10)
- VIII. (a) Derive an expression for log mean temperature difference of parallel flow heat exchanger. How this expression can be modified for counter flow heat exchanger? (5)
- (b) A heat exchanger has 17.5 m² area available for heat transfer. It is used for cooling oil at 200 °C by using water available at 20 °C. The mass flow rate and specific heat of oil are 10,000 kg/hr and 1.9 KJ/kgK and that of water are 3000 kg/hr and 4.187 KJ/kgK. If the overall heat transfer coefficient is 300 W/m²k, estimate the outlet temperatures of oil and water for parallel flow and counter flow arrangements by using NTU method. (10)

OR

- IX. A heat exchanger is required to cool 55,000 kg/hr of alcohol from 66 °C to 40 °C using 40,000 kg/hr of water entering at 5 °C. Calculate (i) exit temperature of water (ii) heat transfer rate (iii) surface area required for (1) parallel flow type (2) counter flow type of heat exchanger. (15)
- Take overall heat transfer coefficient $U = 580 \text{ W/m}^2\text{K}$, $C_{p(\text{alcohol})} = 3760 \text{ J/kg.K}$
 $C_{p(\text{water})} = 4180 \text{ J/kg.K}$.